

Eight International Olympiad, 1966

1966/1

In a mathematical contest, three problems, A , B , C were posed. Among the participants there were 25 students who solved at least one problem each. Of all the contestants who did not solve problem A , the number who solved B was twice the number who solved C . The number of students who solved only problem A was one more than the number of students who solved A and at least one other problem. Of all students who solved just one problem, half did not solve problem A . How many students solved only problem B ?

1966/2

Let a , b , c be the lengths of the sides of a triangle, and α , β , γ , respectively, the angles opposite these sides. Prove that if

$$a + b = \tan \frac{\gamma}{2} (a \tan \alpha + b \tan \beta)$$

the triangle is isosceles.

1966/3

Prove: The sum of the distances of the vertices of a regular tetrahedron from the center of its circumscribed sphere is less than the sum of the distances of these vertices from any other point in space.

1966/4

Prove that for every natural number n , and for every real number $x \neq \frac{k\pi}{2^t}$ ($t = 0, 1, \dots, n$; k any integer) we have

$$\frac{1}{\sin 2x} + \frac{1}{\sin 4x} + \dots + \frac{1}{\sin 2^n x} = \cot x - \cot 2^n x.$$

1966/5

Solve the system of equations

$$|a_2 - a_1|x_2 + |a_1 - a_3|x_3 + |a_1 - a_4|x_4 = 1$$

$$|a_2 - a_1|x_1 + |a_2 - a_3|x_3 + |a_2 - a_4|x_4 = 1$$

$$|a_3 - a_1|x_1 + |a_3 - a_2|x_2 = 1$$

$$|a_4 - a_1|x_1 + |a_4 - a_2|x_2 + |a_4 - a_3|x_3 = 1$$

where a_1, a_2, a_3, a_4 are four different real numbers.

1966/6

In the interior of sides BC, CA, AB of triangle ABC , any points K, L, M , respectively, are selected. Prove that the area of at least one of the triangles AML, BKM, CLK is less than or equal to one quarter of the area of triangle ABC .